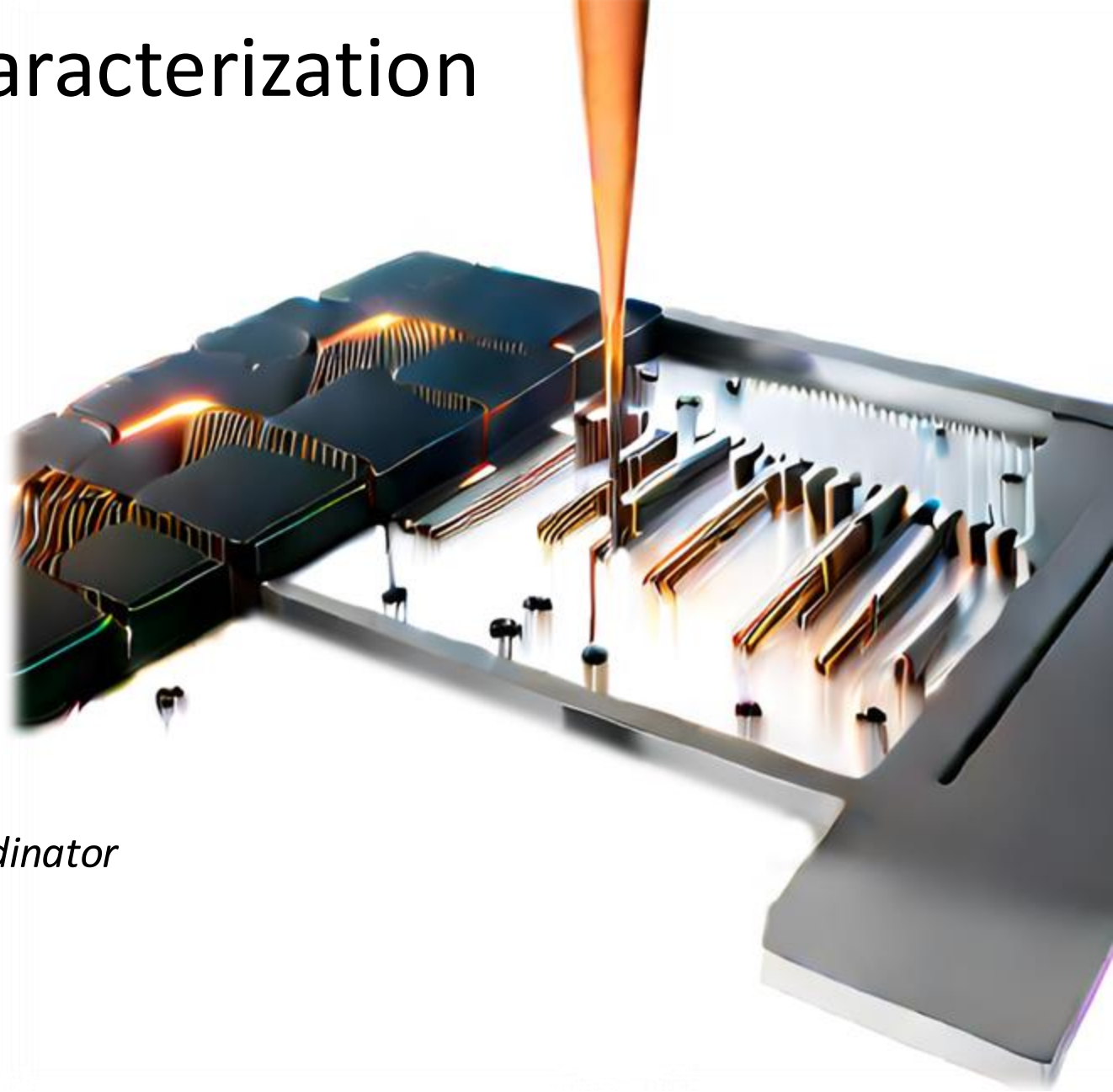


68320 Nanofabrication and Nanocharacterization Techniques



UNIVERSITY OF
TECHNOLOGY SYDNEY

Lecture 1 Introduction



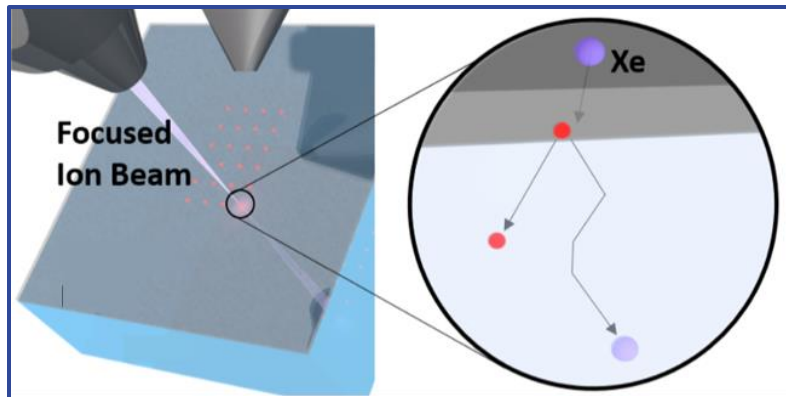
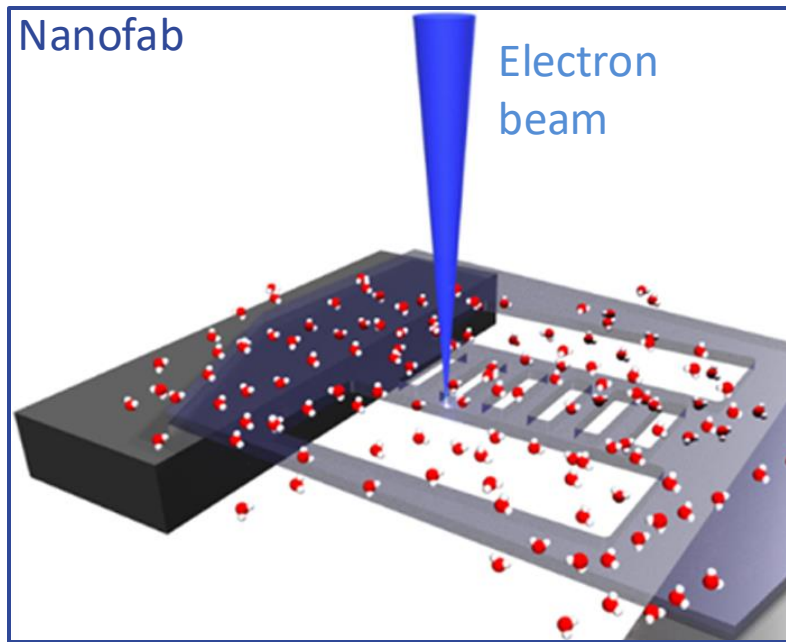
Lecturer & Subject Coordinator

Prof Milos Toth

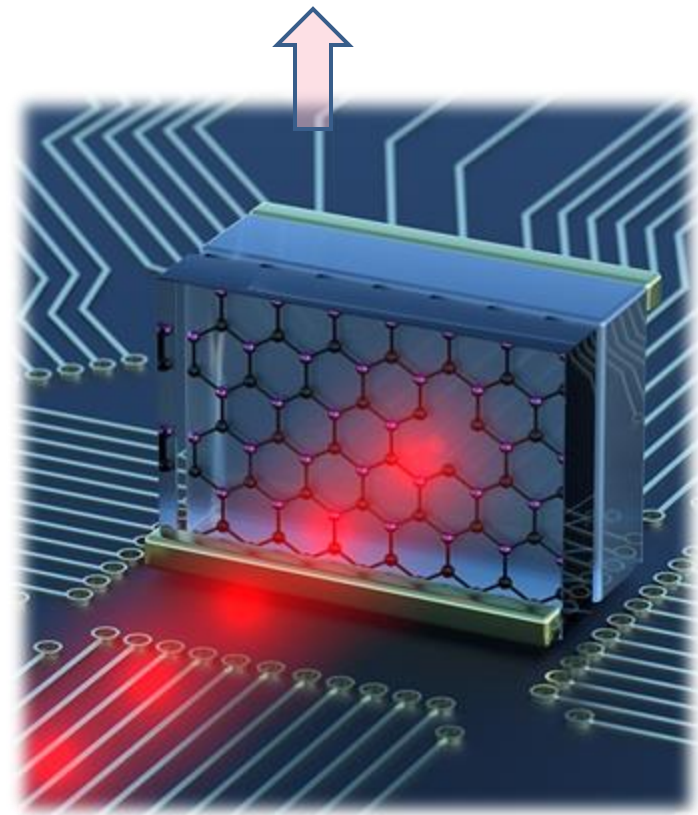
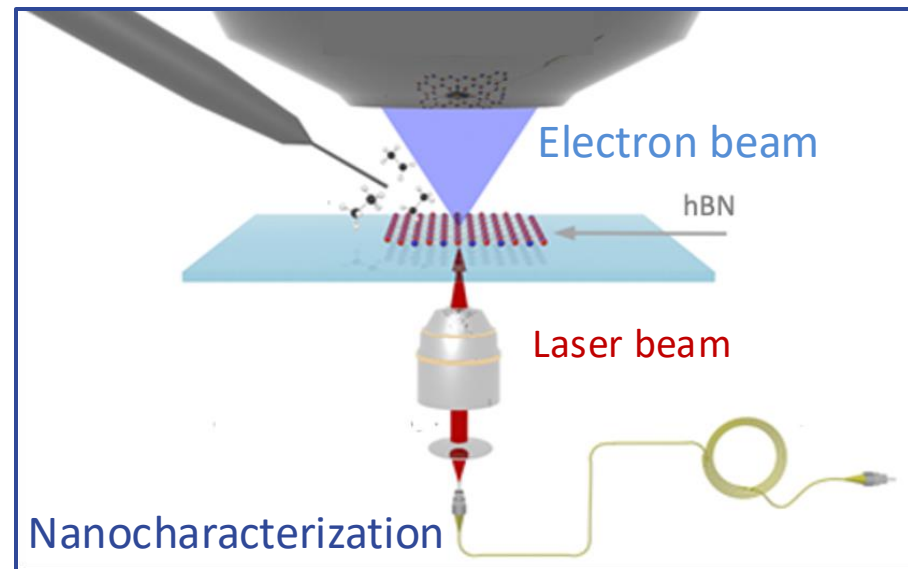
milos.toth@uts.edu.au

Brief introduction to electron, ion and laser beam techniques: imaging, analysis and nanofabrication

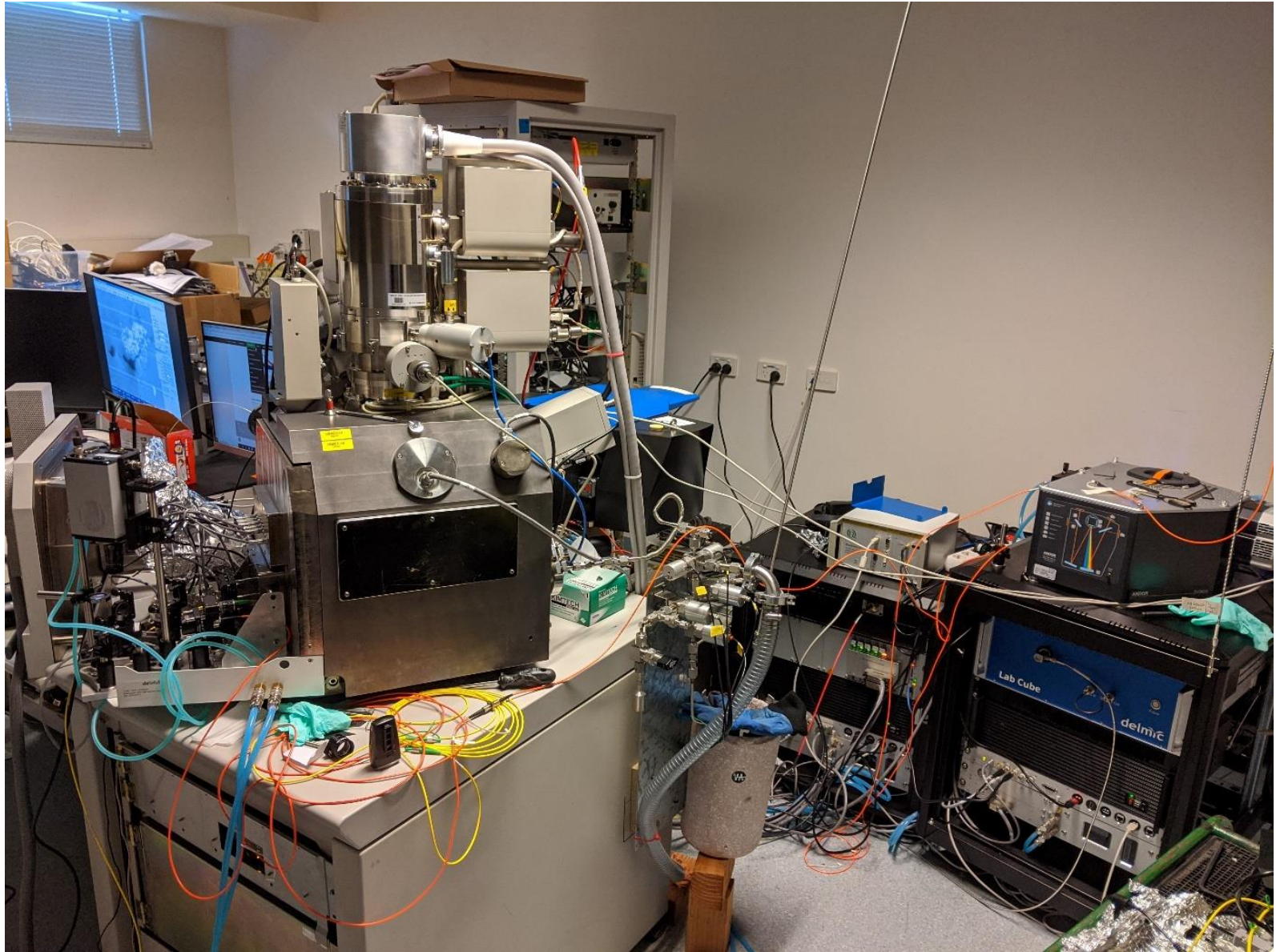
Overview



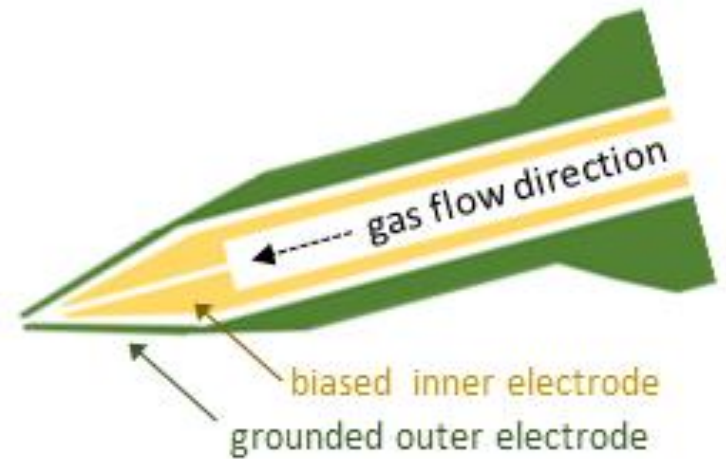
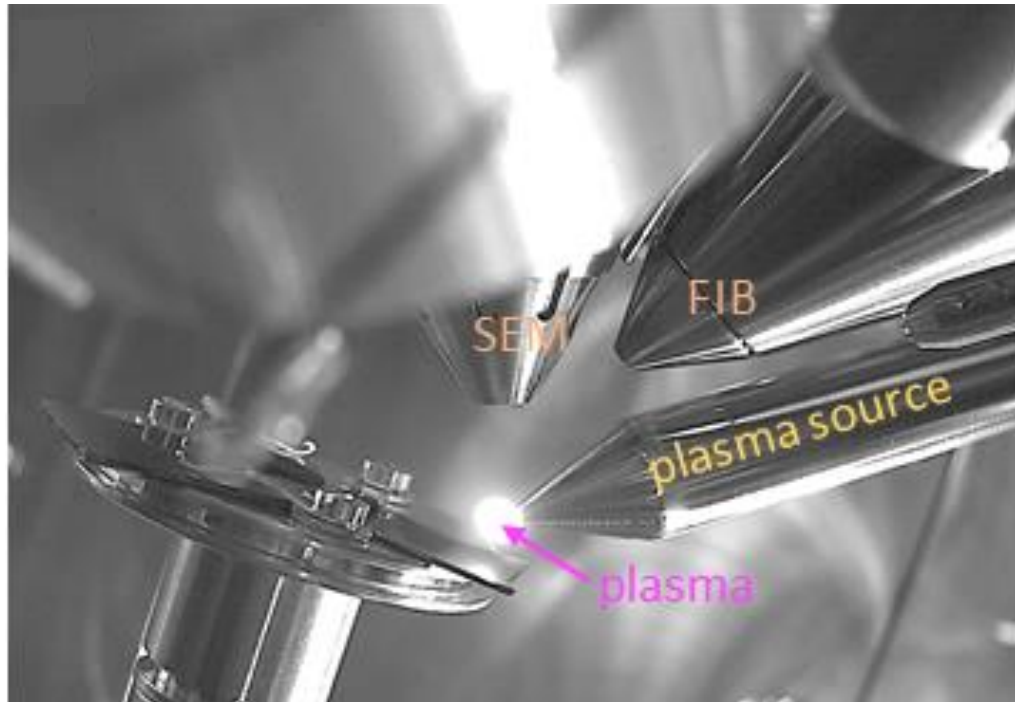
Material processing: Optical doping



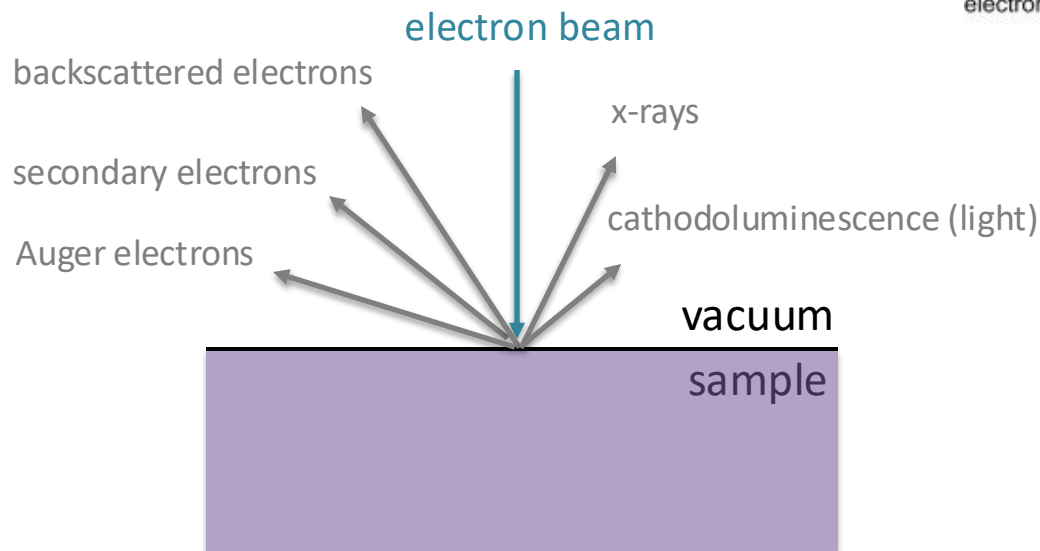
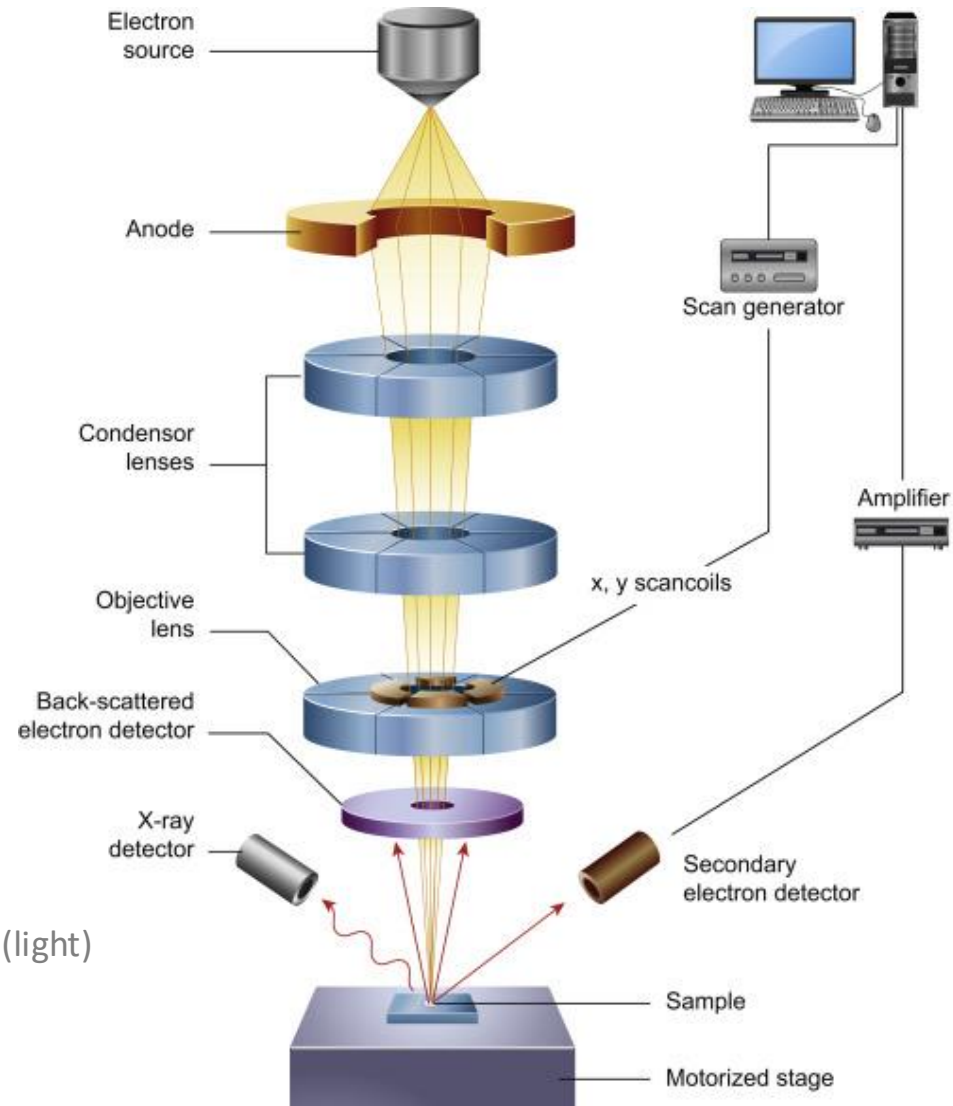
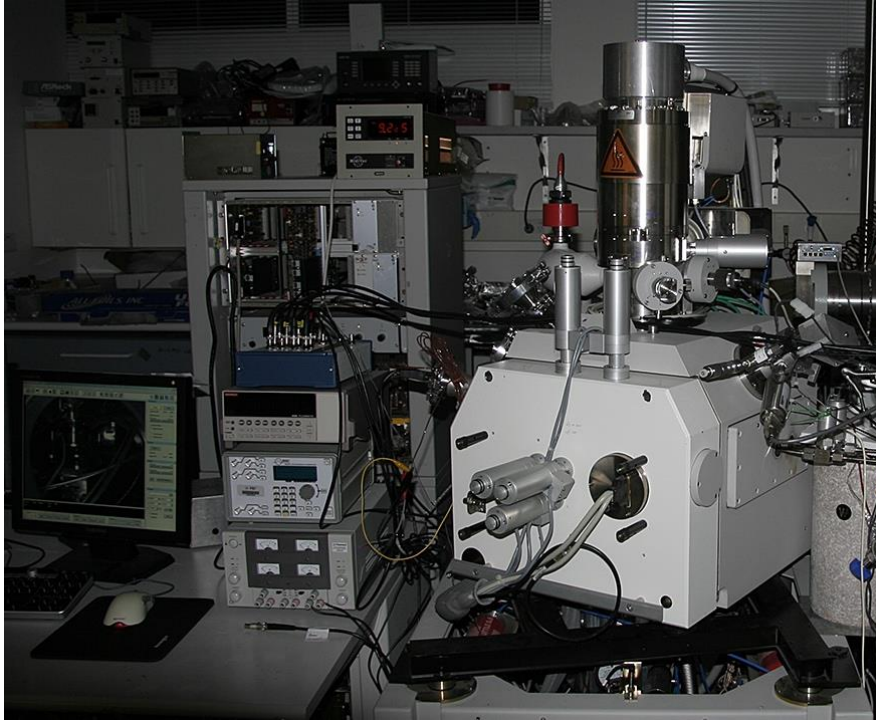
Lab: SEM, FIB, CL, PL



Lab: Inside one of these microscopes

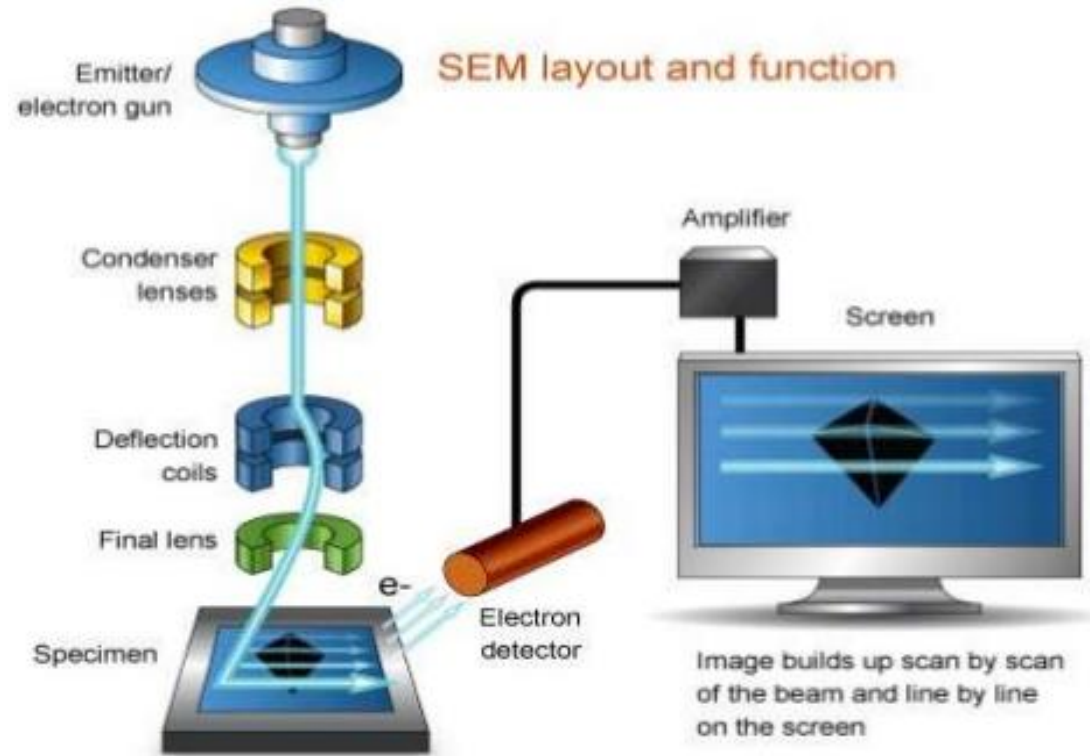
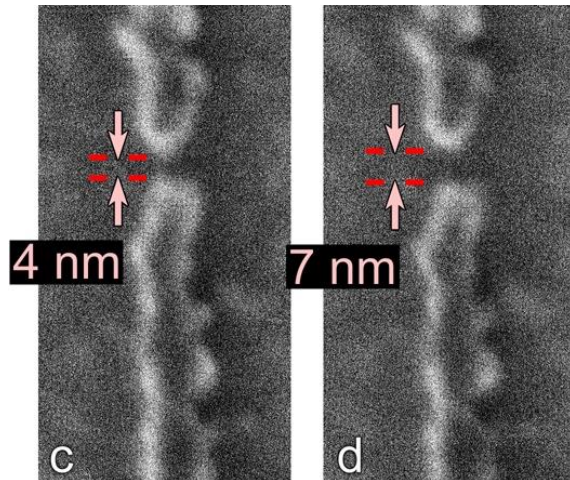
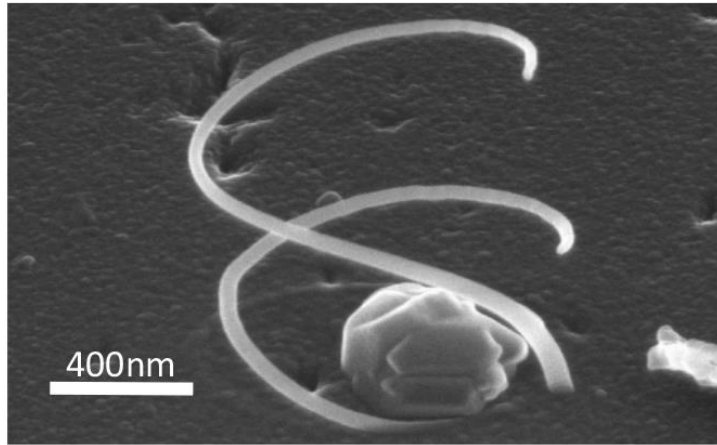


Intro: Scanning electron microscope

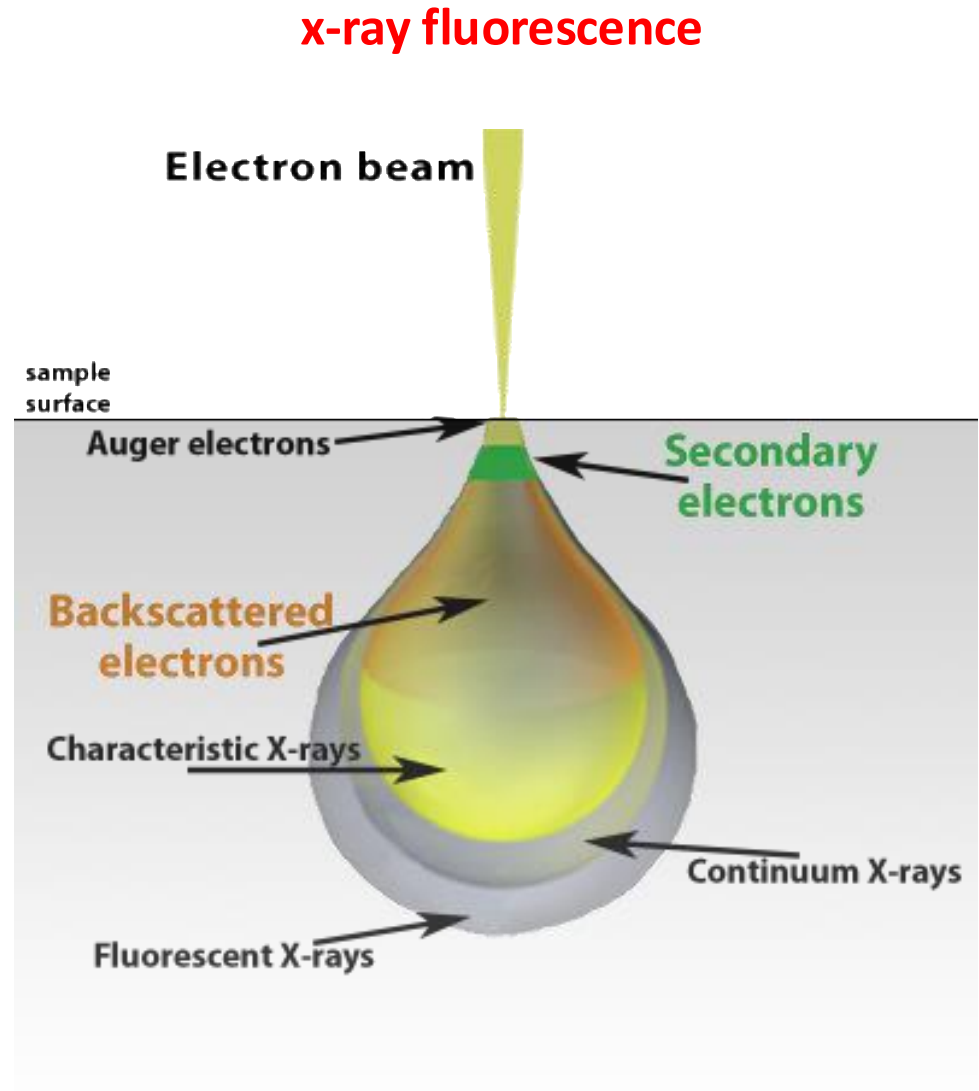
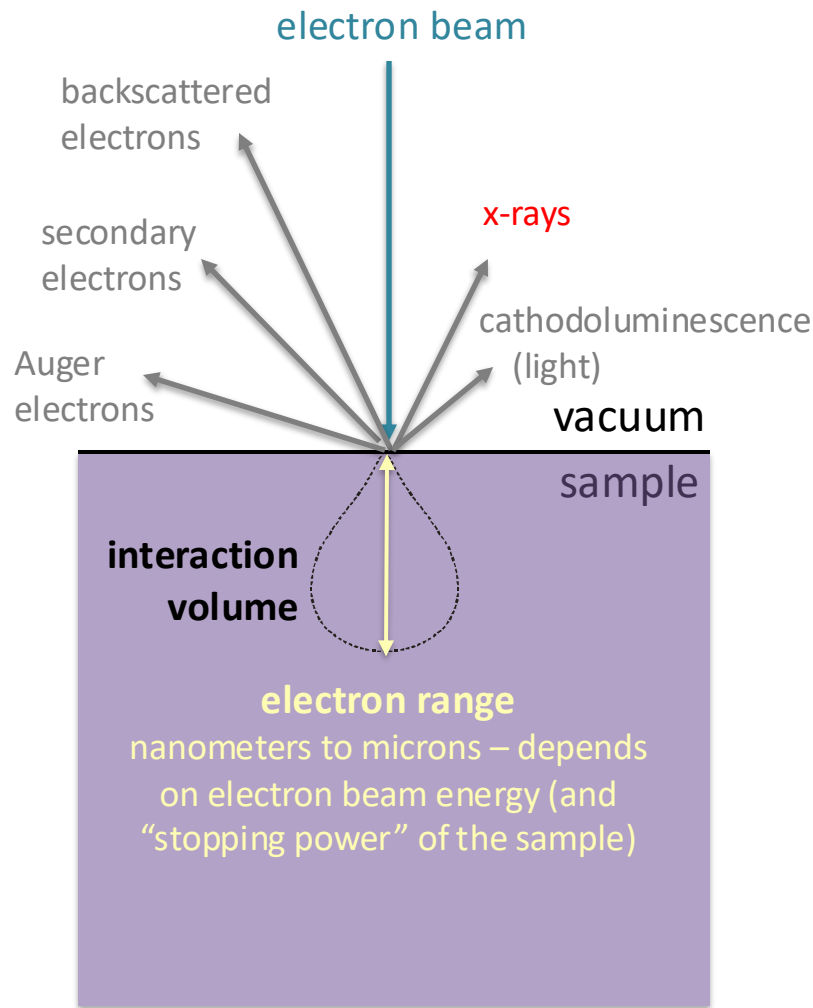


Intro: SEM image formation

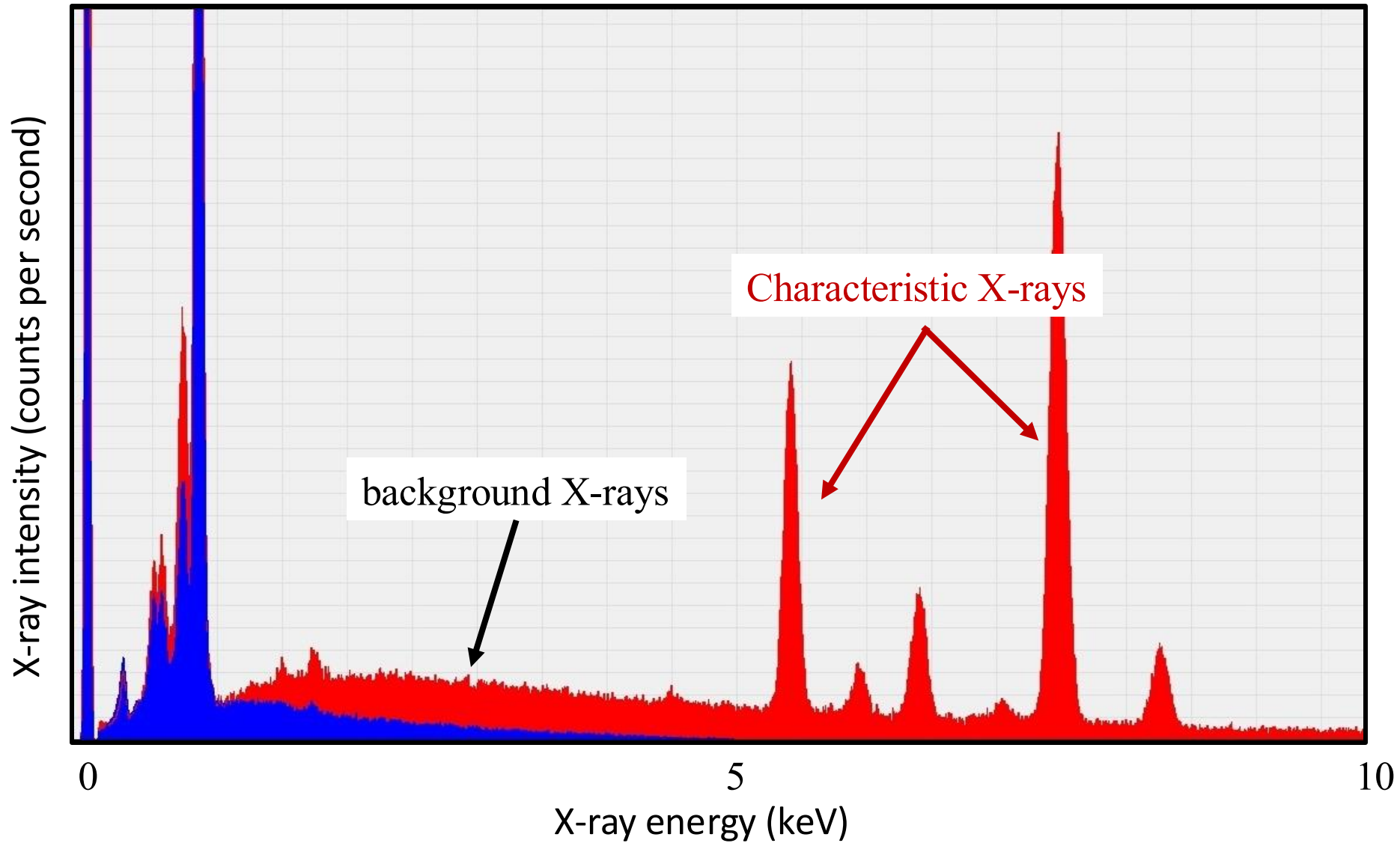
secondary electron images



SEM: imaging/analysis signals

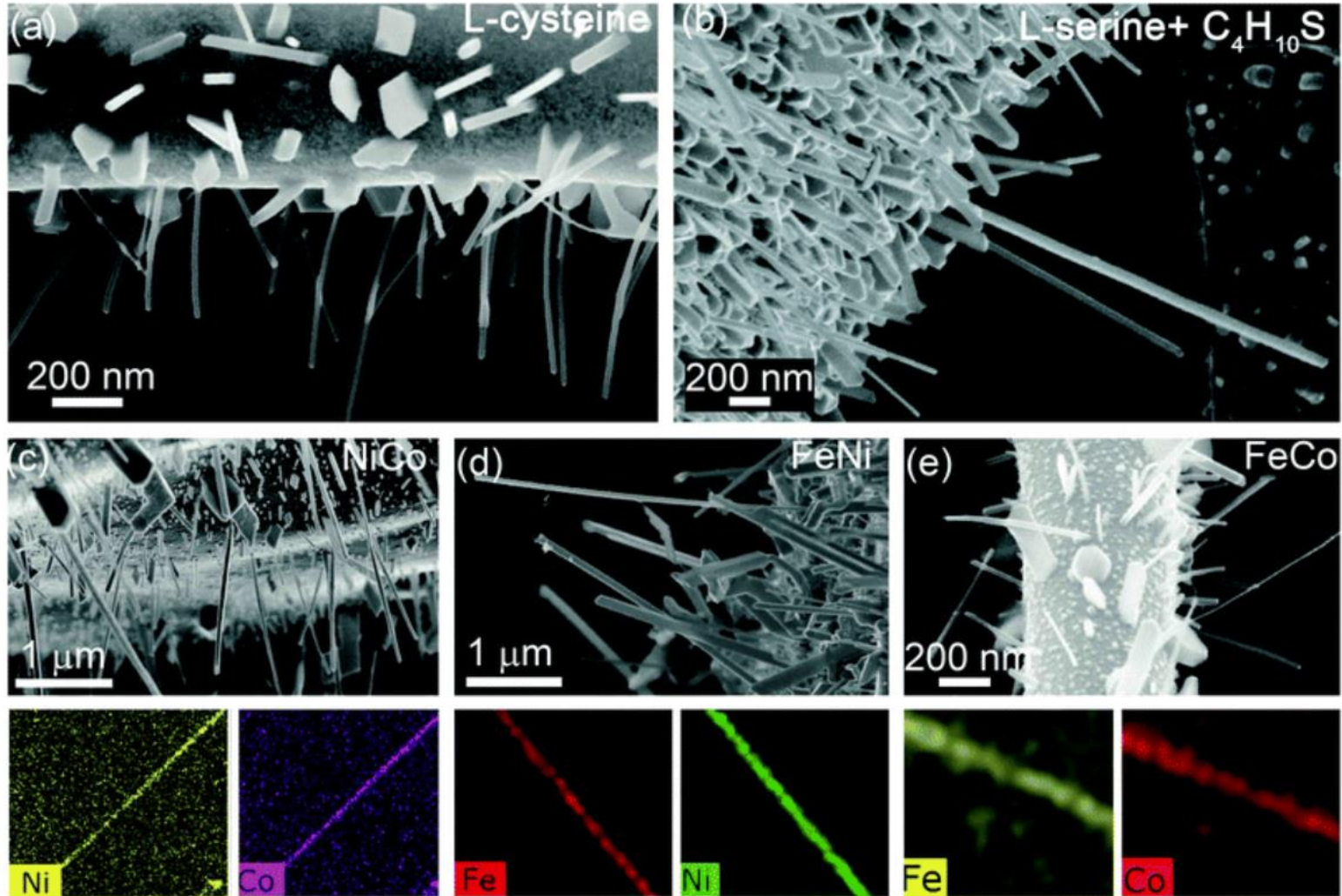


X-ray fluorescence: Spectroscopy



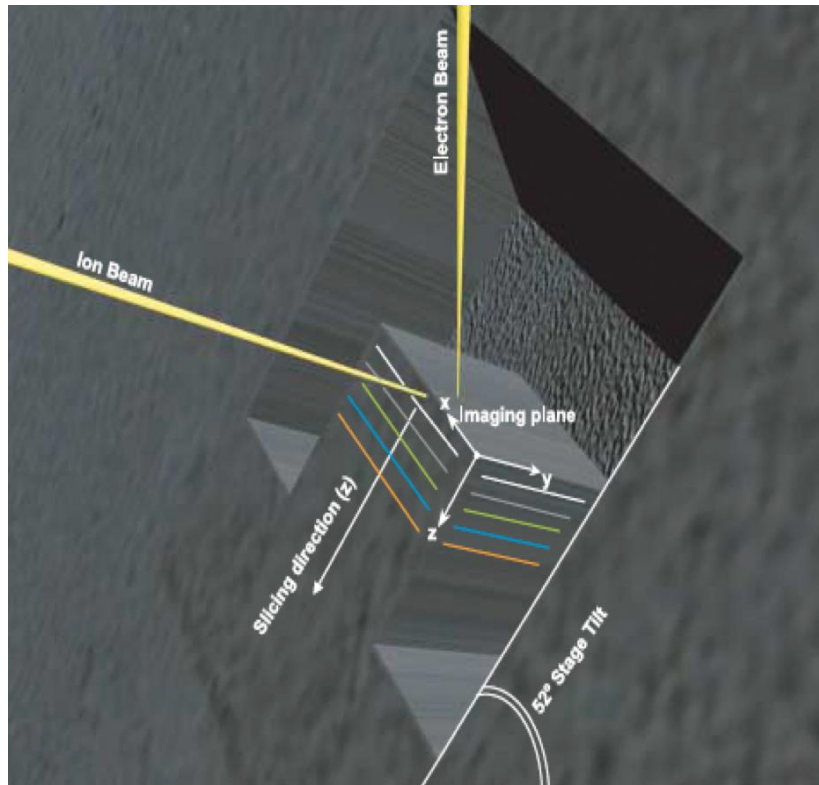
X-ray imaging (of nanowires)

SEM images show nanowire morphology

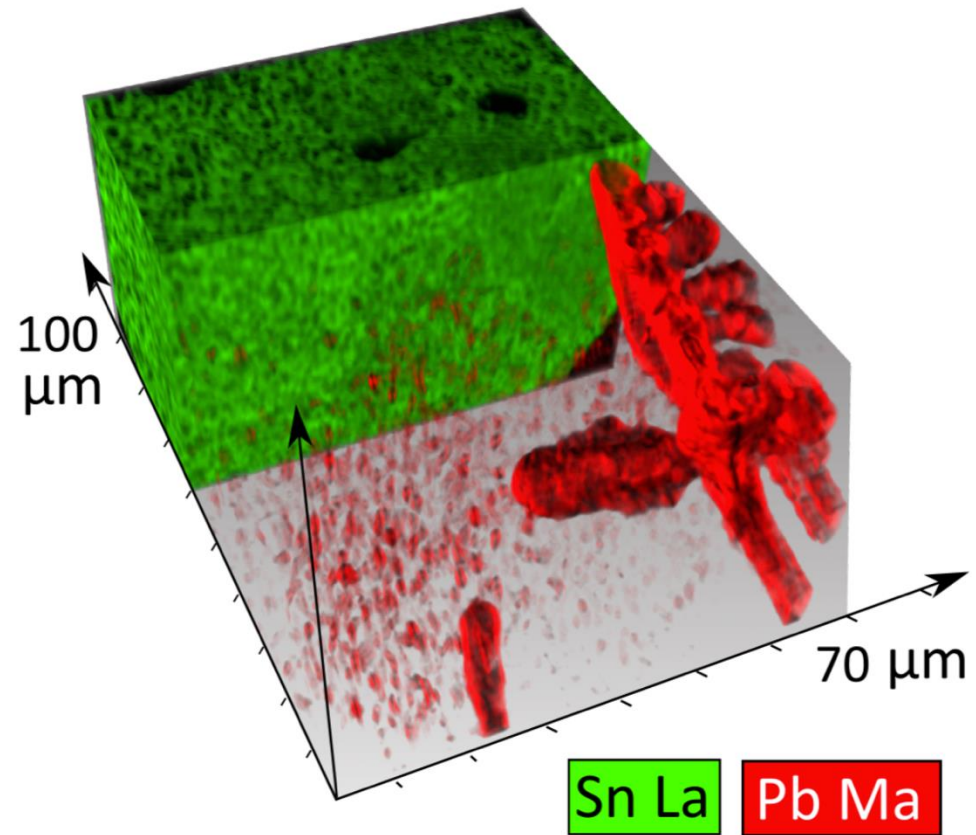


X-ray maps reveal elemental composition of individual nanowires

3D SEM & x-ray analysis: Focused ion beam (FIB) microscopy



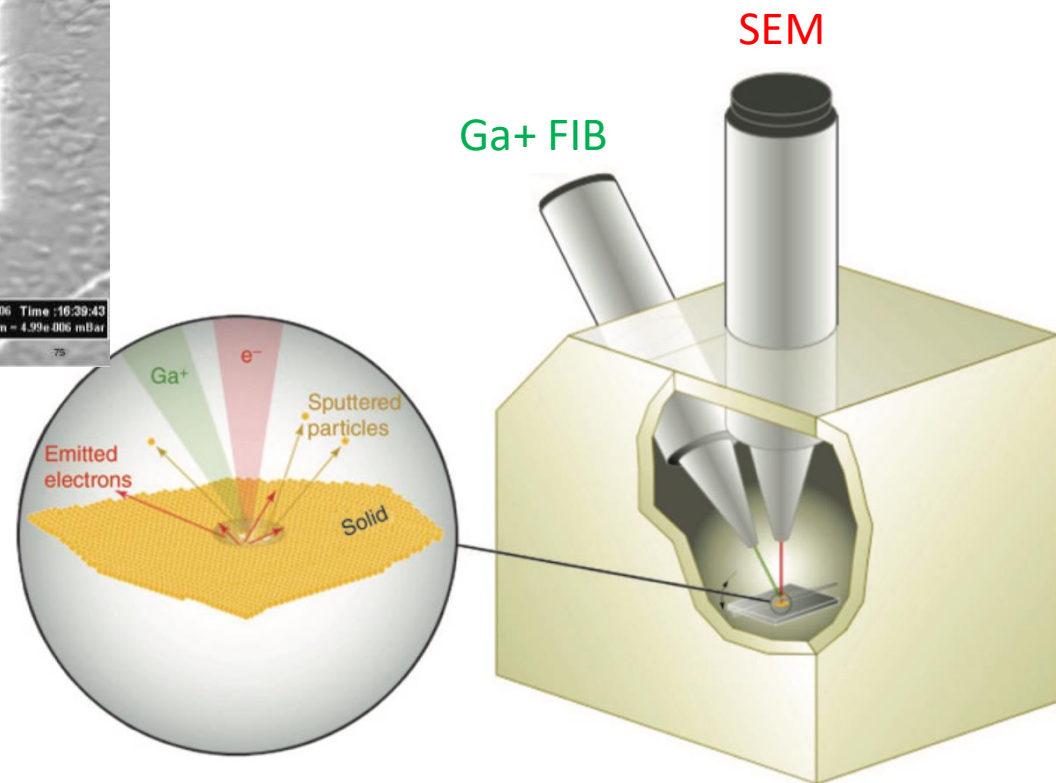
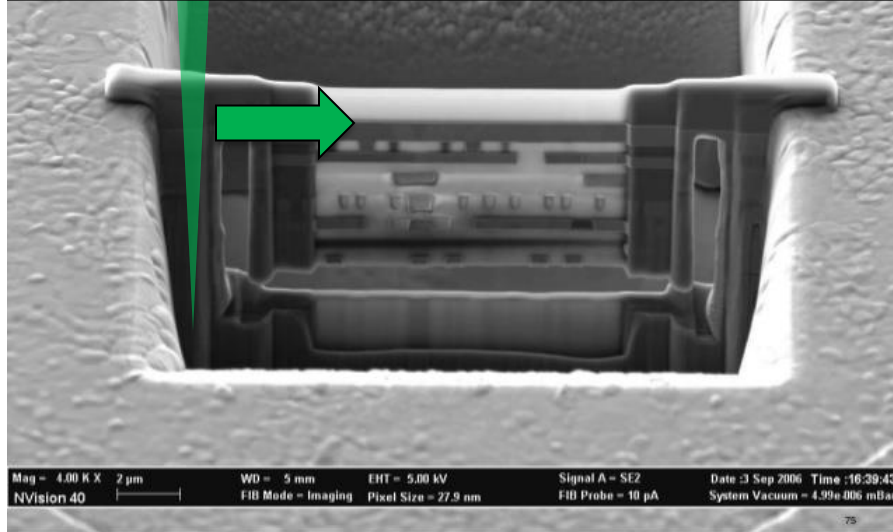
L. HOLZER, et al., *Journal of Microscopy*
Vol. 216, Pt 1 October 2004, pp. 84–95



3D EDS reconstruction of
 $\text{Sn}_{60}\text{Pb}_{40}$ solder (TESCAN)

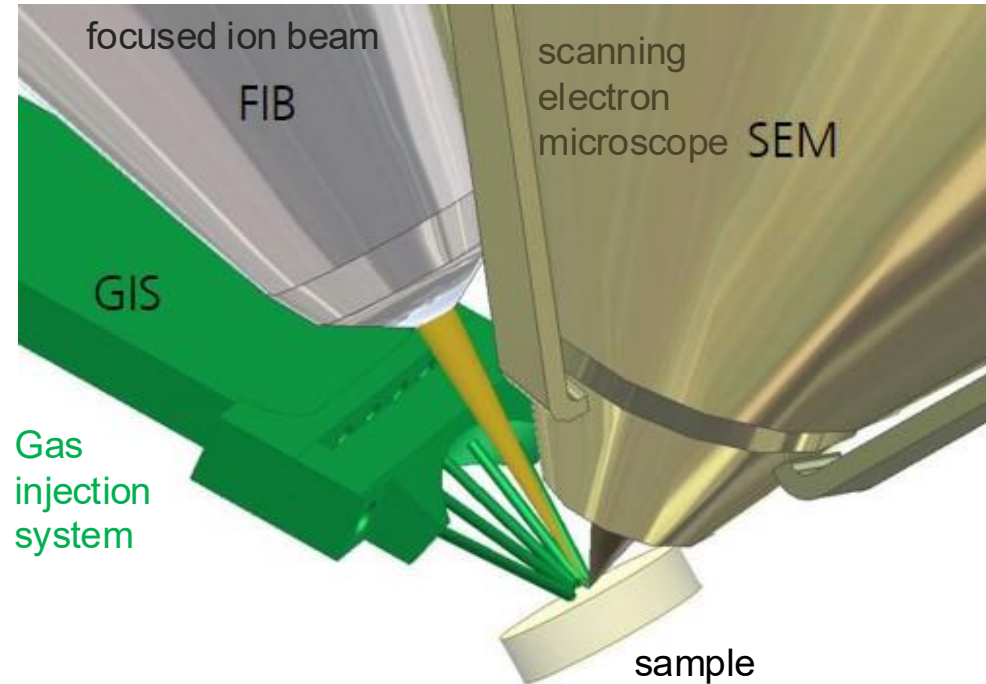
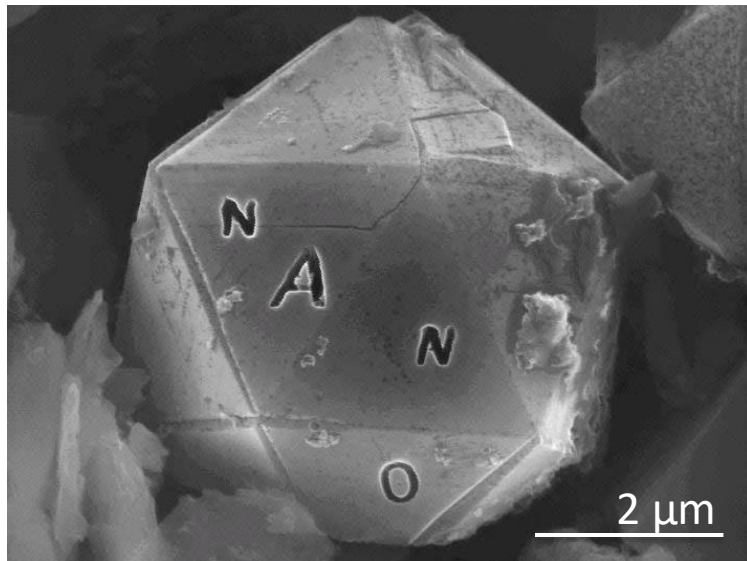
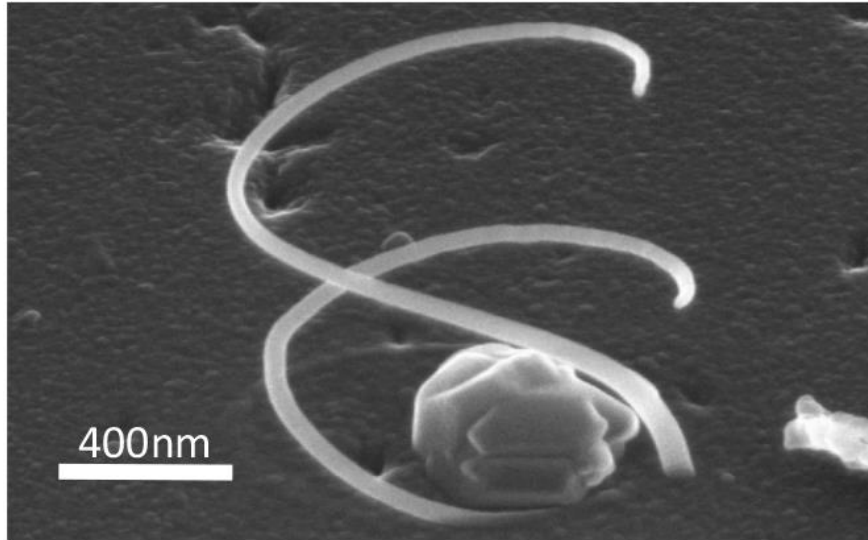
SEM of cross-sections fabricated using a focused ion beam (FIB)

SEM image: cross-section of a microchip
fabricated using a FIB, imaged by SEM



Gas-assisted nanofabrication

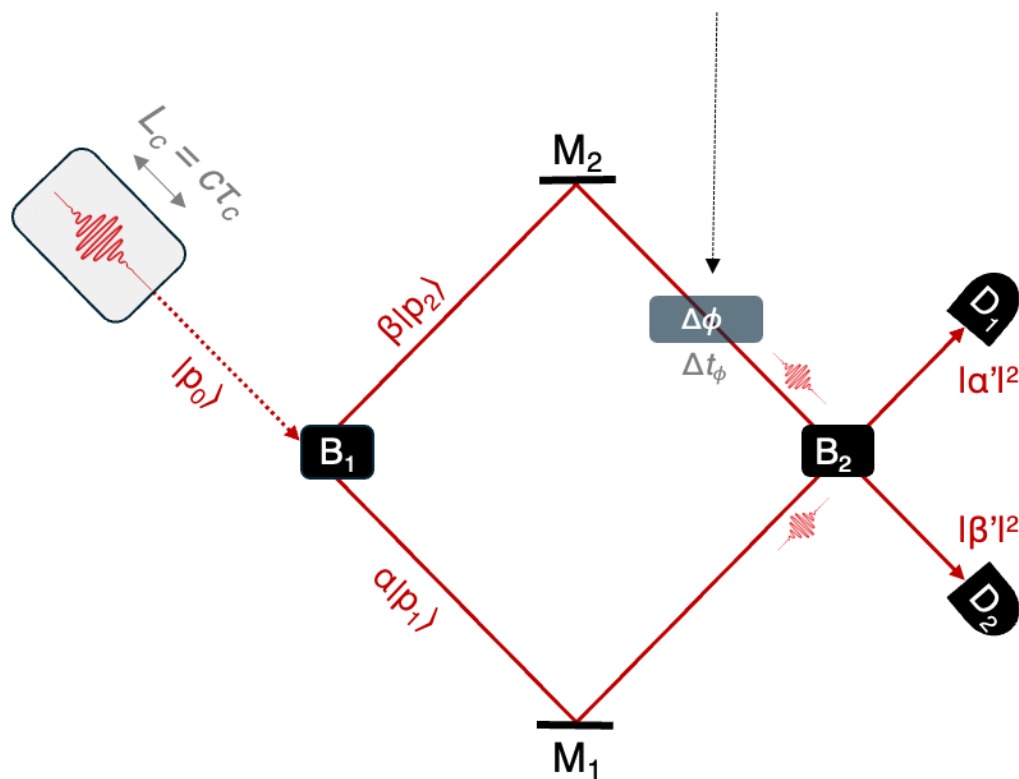
3D spirals grown by a (slowly)
scanned electron beam



"NANO" etched into a microdiamond
by an electron beam

Single photon quantum microscopy

the sample is the
'phase shifter'



Changes in interference
caused by the sample are
the imaging signal.

i.e., very subtle
interactions between a
photon in a superposition
state & a sample that
modify its phase (but do
not collapse the wave
function) generate a
signal and hence 'phase
contrast' in images

End of “brief introduction”/teaser

Timetable & assessments

Check out everything on Canvas, particularly:

- [Subject information \(includes assessment info\)](#)
- [Modules > Get started](#)
- [Assignments > Assignments will be rolled out here](#)
- [Quizzes > Class tests will be done online \(but onsite at UTS\)](#)

UTS

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68320 Nanofabrication and Nanocharacterization Techniq... > Modules

Spring 2025 (City campus)

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Your teaching staff

How to be successful in this subject

Timetable 2025.pdf

Assessment overview

Get started with UTS Library

Study help and support

Important guidance

Subject schedule

Calendar/timetable

Lectures (Wed, 8:30 AM, CB11.04.205)

30-Jul	1	Introduction
6-Aug	2	Electron optics and electron-solid interactions
13-Aug	3	Focused ion beam microscopy and nanofabrication
20-Aug	-	Revision / Q&A (test preparation)
27-Aug	4	Emission of cathodoluminescence, x-rays & electrons
3-Sep	5	Quantum microscopy
10-Sep	6	Transmission Electron Microscopy
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24-Sep	-	STUVAC
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8-Oct	8	Advanced nanofabrication & nanocharacterization 2
15-Oct	-	Revision / Q&A (test preparation)

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29-Aug	4	Monte Carlo simulations of cathodoluminescence
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19-Sep	-	Class Test 2 in place of tute
26-Sep	0	No tute (STUVAC)
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10-Oct	6	Advanced analysis of nanocharacterisation data
17-Oct	-	Class Test 3 in place of tute

Pracs (Thurs, 1 PM, CB04.02.310, "MAU lab")

31-Jul	0	Introduction to electron and ion beam systems
7-Aug	1	Scanning electron microscopy
14-Aug	2	Focused ion beam microscopy and nanofabrication
21-Aug	-	No prac this week (class test)
28-Aug	3	X-ray analysis
4-Sep	4	Advanced nanofabrication and nanocharacterisation

I will send an email at the start of each week with a reminder of what is scheduled.

Assessments

- 50% for Practical and Tutorial Work submitted in the form of **3 portfolios** throughout the semester
- 50% for Class Tests held on 3 dates throughout semester

The best way to prepare is to attend lectures and participate in discussions – I will ask many questions that are representative of what will be in the tests.

Class Test dates are in the timetable. Test will be done online, in the same computer room as tutorials (you must be onsite at UTS to do a class test).

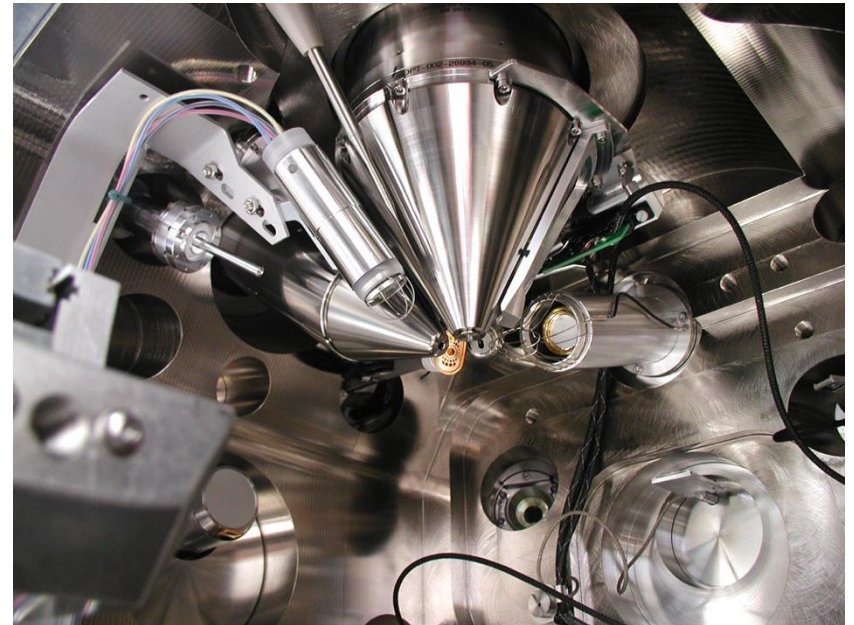
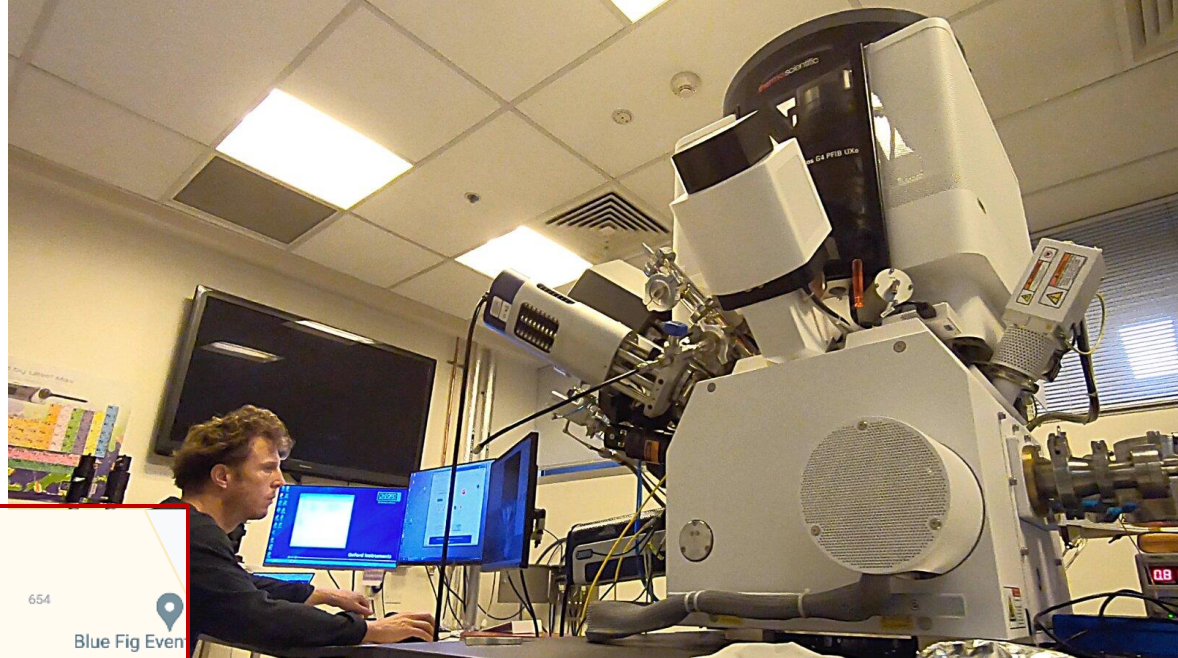
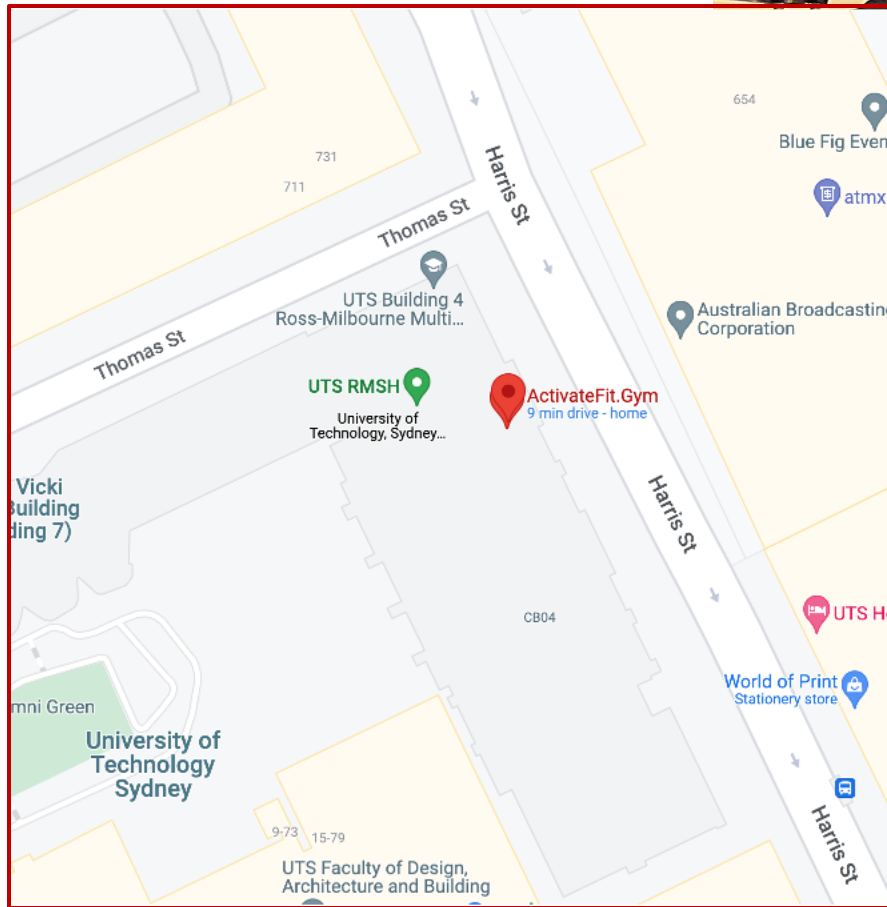
Tute & Prac portfolio due dates:

- Portfolios 1 & 2: Friday 12 Sept (midnight)
- Portfolio 3: Friday 24 Oct (midnight)

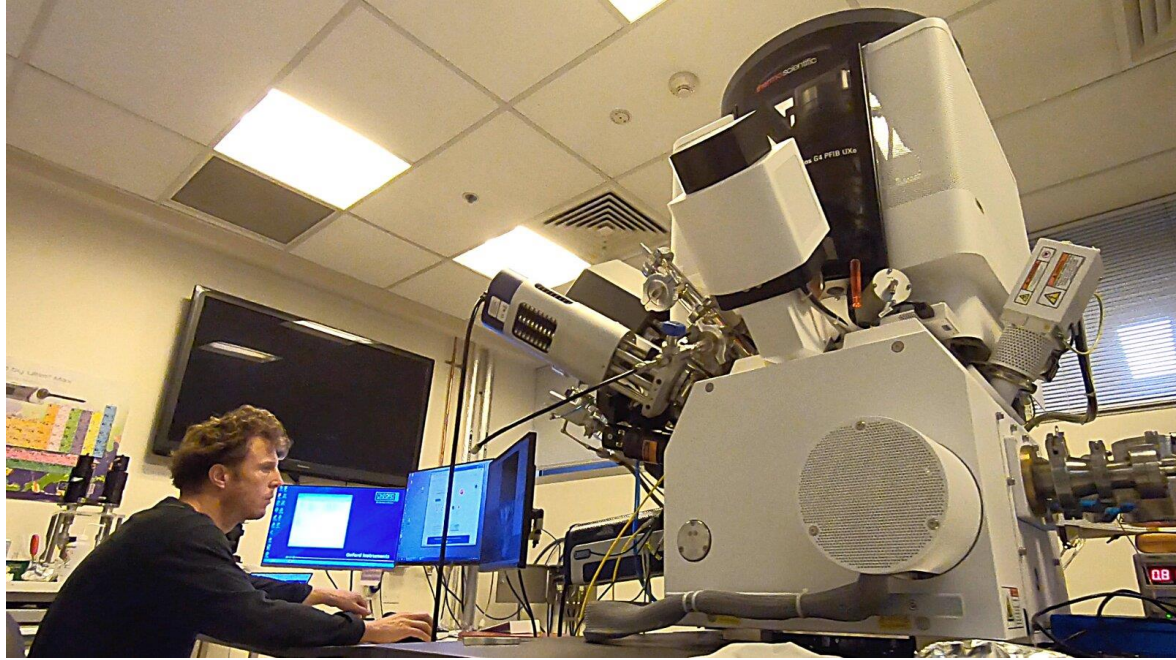
Labs

CB04.02.310

Corner of Harris & Thomas St
Near the UTS gym

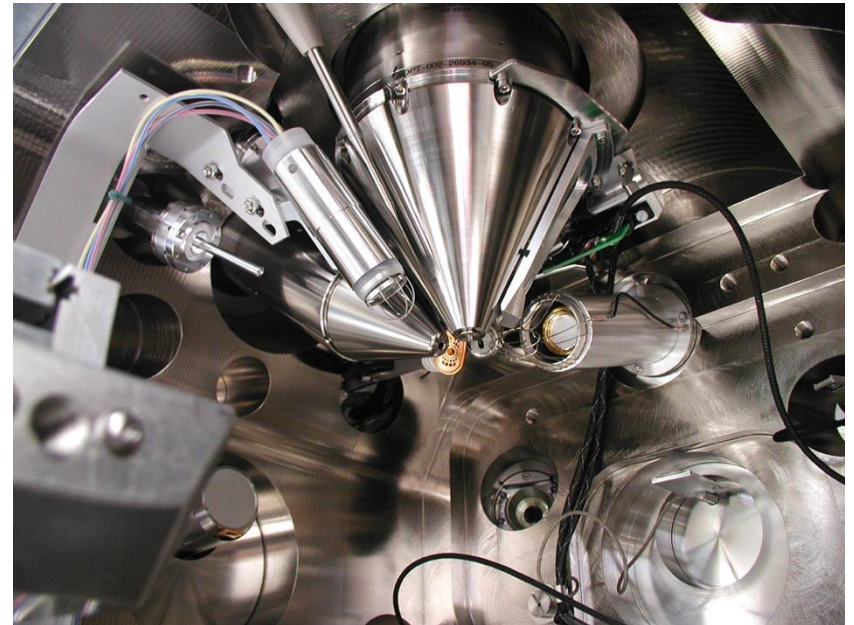


Labs



Class will be split into 2 groups:

- 1 – 2 PM (Thurs)
- 2 – 3 PM (Thurs)



Reading material [Canvas > Modules > LITERATURE]

Spring 2025 (City campus)

Home Subject Information Reading List Announcements People Modules Assignments Discussions Grades Quizzes Files Rubrics Pages

Get started

Honours Info

LITERATURE

Start with these 3 papers

Spring 2023 (City campus)

LITERATURE

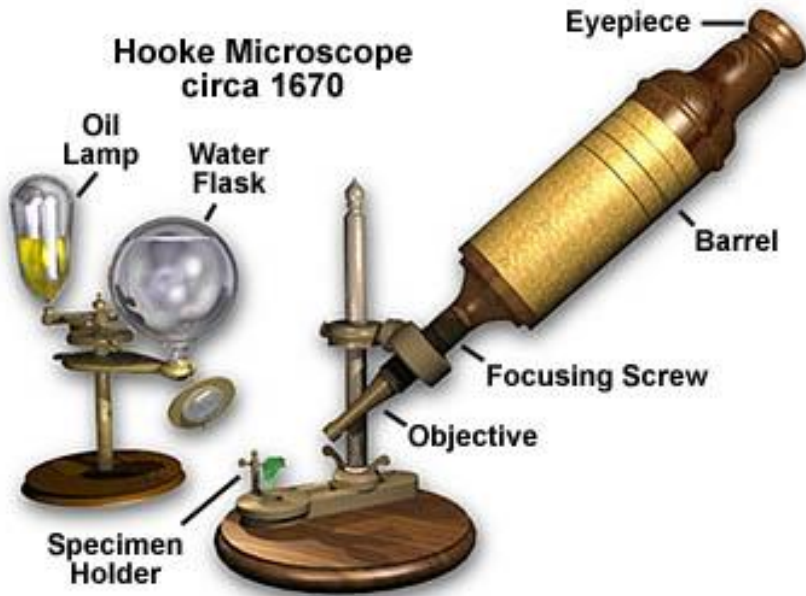
The following articles are provided as PDFs, and will be referred to as follows:

- 01A. FIB1.pdf
- 01B. FIB2.pdf
- 02. SEM - Low Voltage.pdf
- 03. cathodoluminescence.pdf
- 04. FIB and 3D SEM imaging.pdf
- 05. EDX Mapping.pdf
- 06. SEM Resolution.pdf
- 07. CASINO - intro.pdf
- 08. CASINO - 3D.pdf
- 09. CASINO - elastic.pdf
- 10. CASINO - inelastic.pdf
- 11. Gas-assisted nanofab - SEM and FIB.pdf
- 12. Models of e-beam nanofab.pdf
- 13. self-assembly.pdf
- 14. CL Depth profiling.pdf
- 15. Recoil Implantation.pdf
- 16. Recoil Implantation 2 - gas.pdf

Prep for tutorial 1

Microscopy

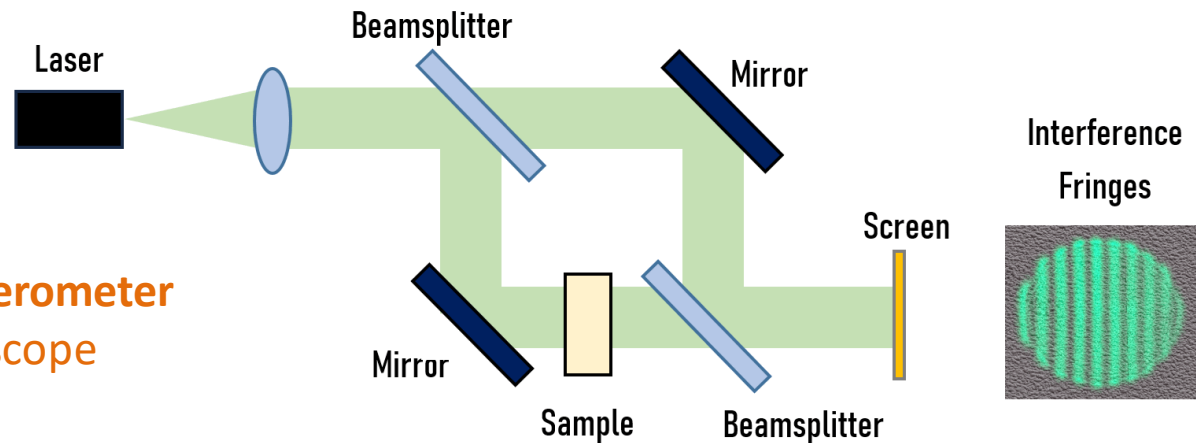
First Microscope (magnification $\sim 270\times$)



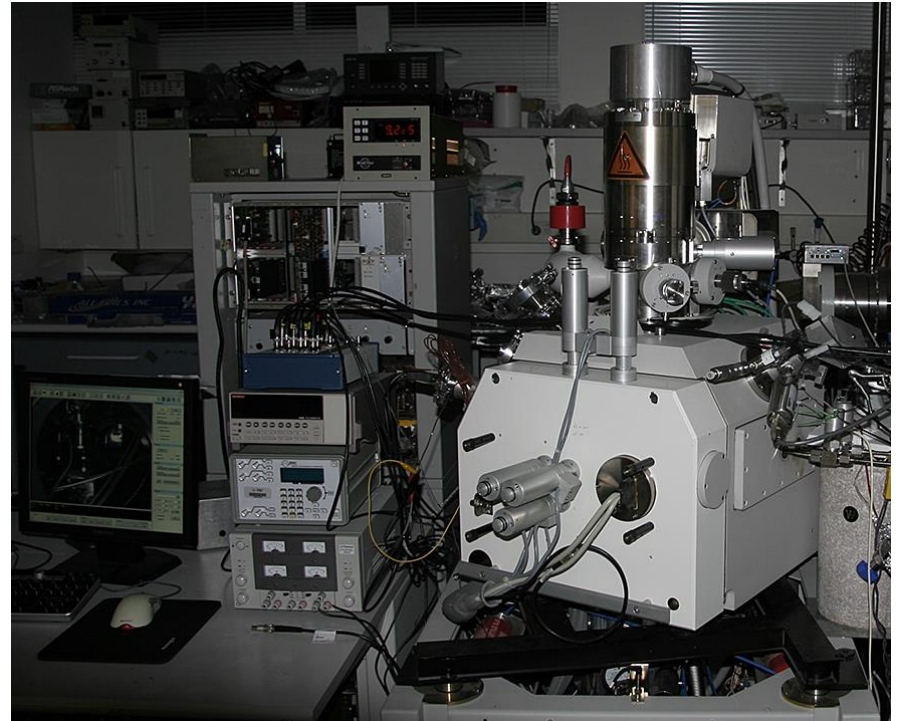
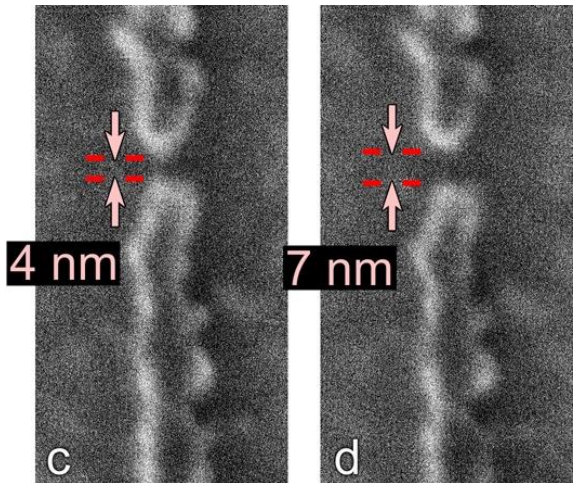
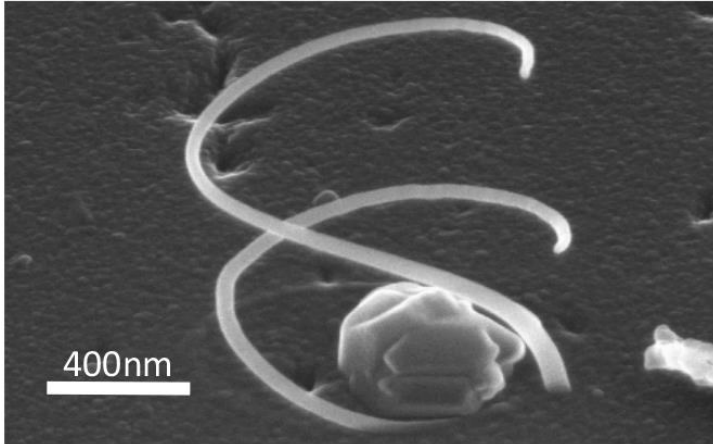
Transmission electron microscope (magnification $>10,000,000\times$)



Mach-Zehnder interferometer phase contrast microscope

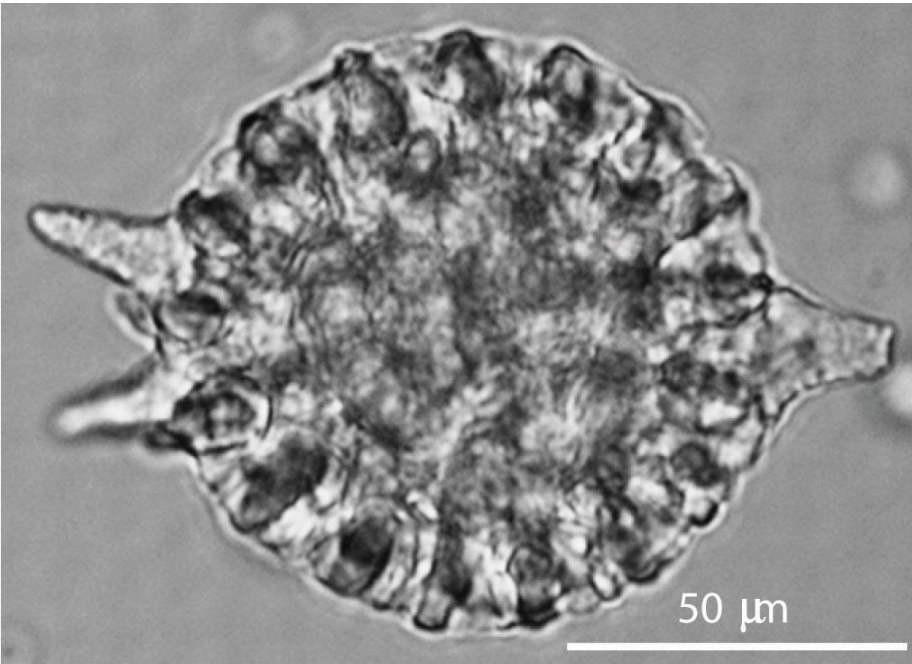


Scanning electron microscopy



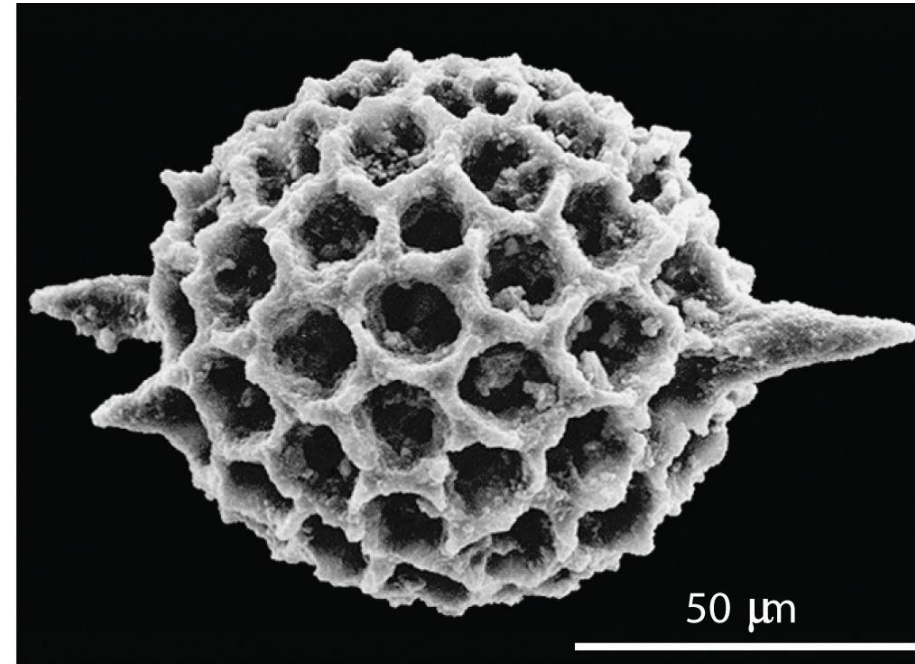
Optical/light vs SEM/electrons

Light Microscope



(a) Radiolarian under light microscope

Scanning Electron Microscope (SEM)



(b) Radiolarian under electron microscope

SEM: Improved “spatial detail”, sharper edges and contrast, larger depth of field

The Radiolaria, also called Radiozoa, are protozoa of diameter 0.1–0.2 mm that produce intricate mineral skeletons, typically made of silica, and contain a central capsule dividing the cell into the inner and outer portions of endoplasm and ectoplasm.

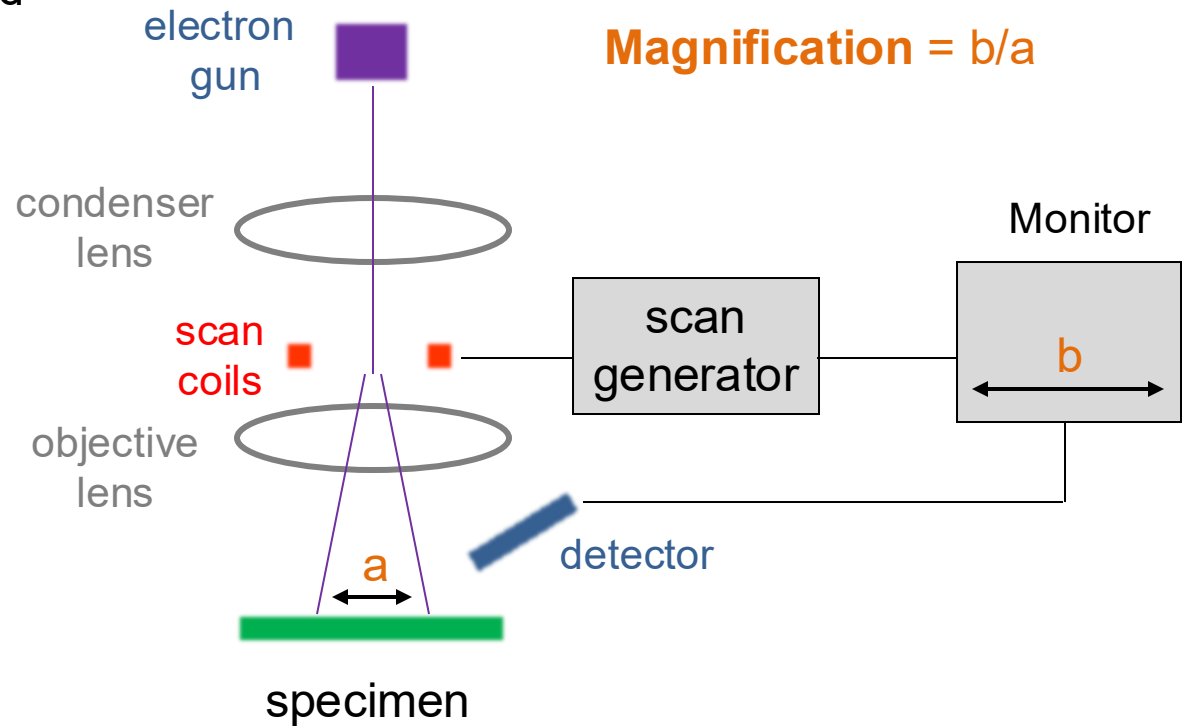
SEM basics & magnification

all SEMs operate by irradiating the specimen with a finely focussed beam of electrons, producing emission signals (SE, BSE, X-ray, etc.) that can be used to form an image

Electron Gun: used to produce electrons, pre-focus beam and accelerate to 1 - 30 keV

Electron Lens: used to provide a focussed, de-magnified spot at the specimen plane

Imaging System: used to scan the beam, achieve magnification, and produce image contrast



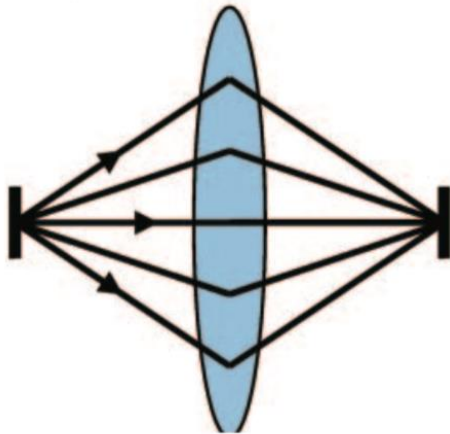
What limits spatial resolution?

Resolution is limited by **aberrations** and by **diffraction** that cause blurring of images

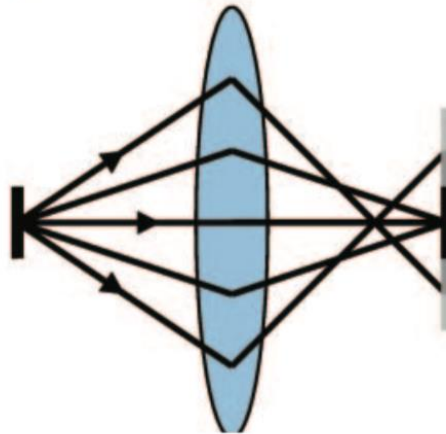


caused by imperfections of the imaging system
2 examples:

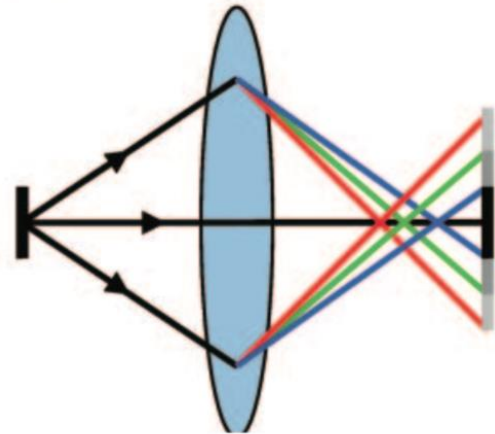
a perfect lens



b spherical aberration

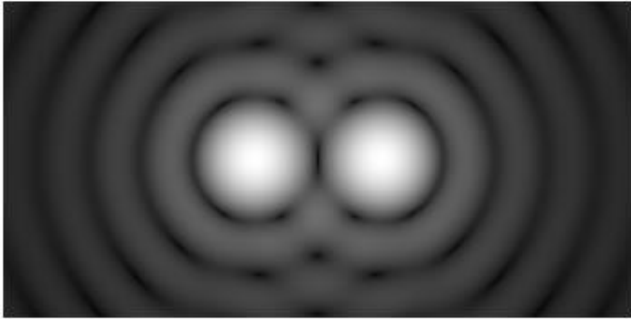


c chromatic aberration

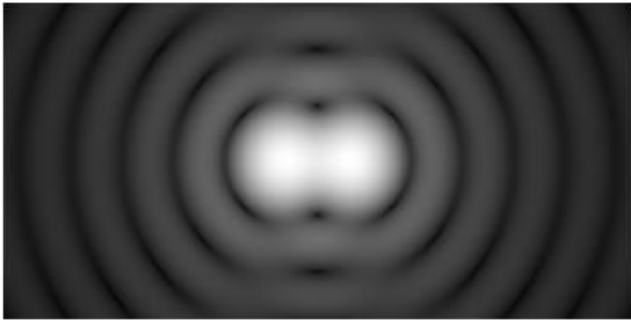


Diffraction aberration

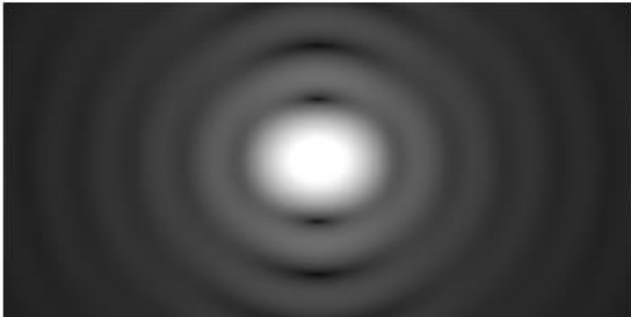
fundamental resolution limit imposed by the wave nature of particles (photons ,electrons and all other particles/objects)



2 objects far apart – well resolved



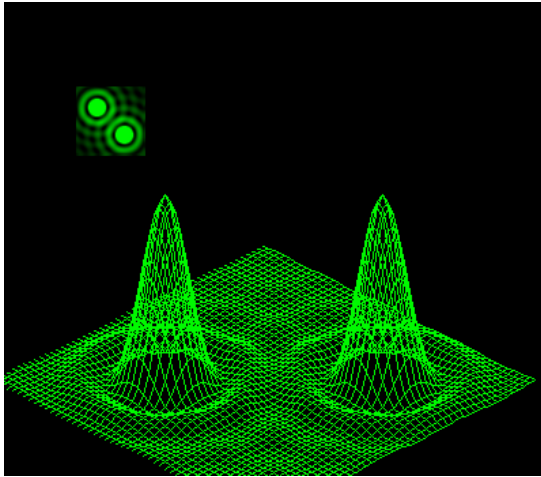
2 objects at the diffraction limit of resolution defined by the Rayleigh criterion



2 objects close together – unresolved

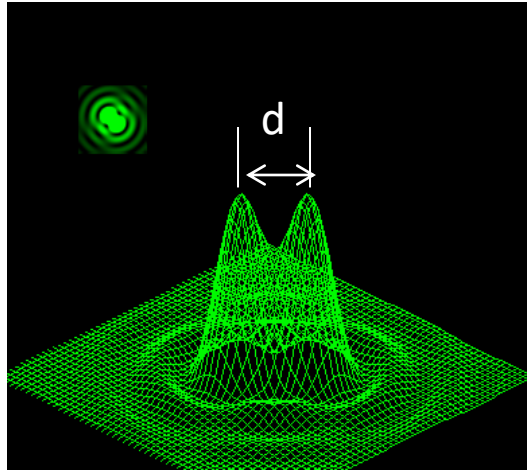
What is resolution?

resolved

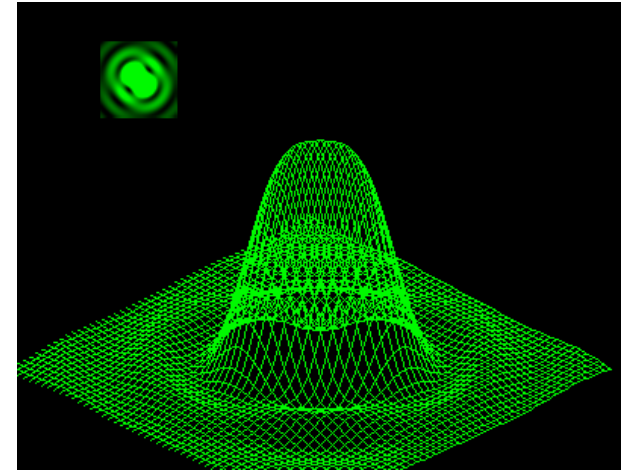


diffraction-limited image of 2 features much smaller than the resolution

resolved



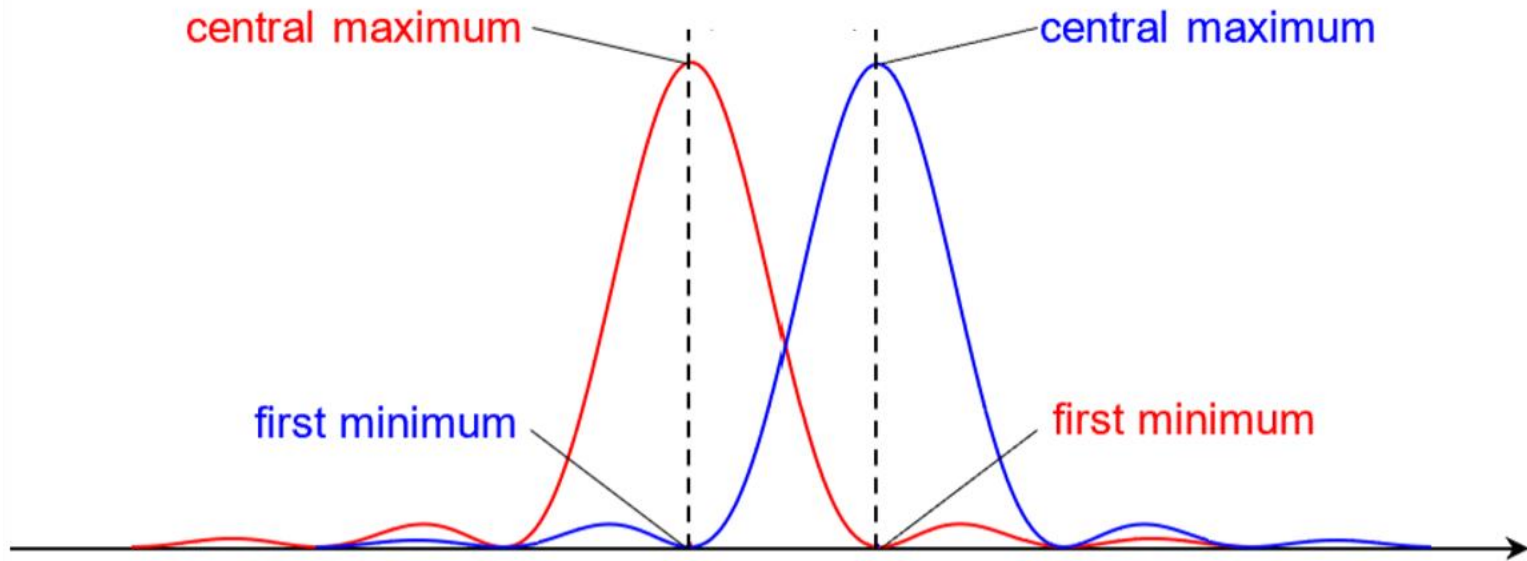
unresolved



is this an image of 1 broad feature or 2 small features?

spatial resolution: smallest distance between two features that can be distinguished as two separate entities

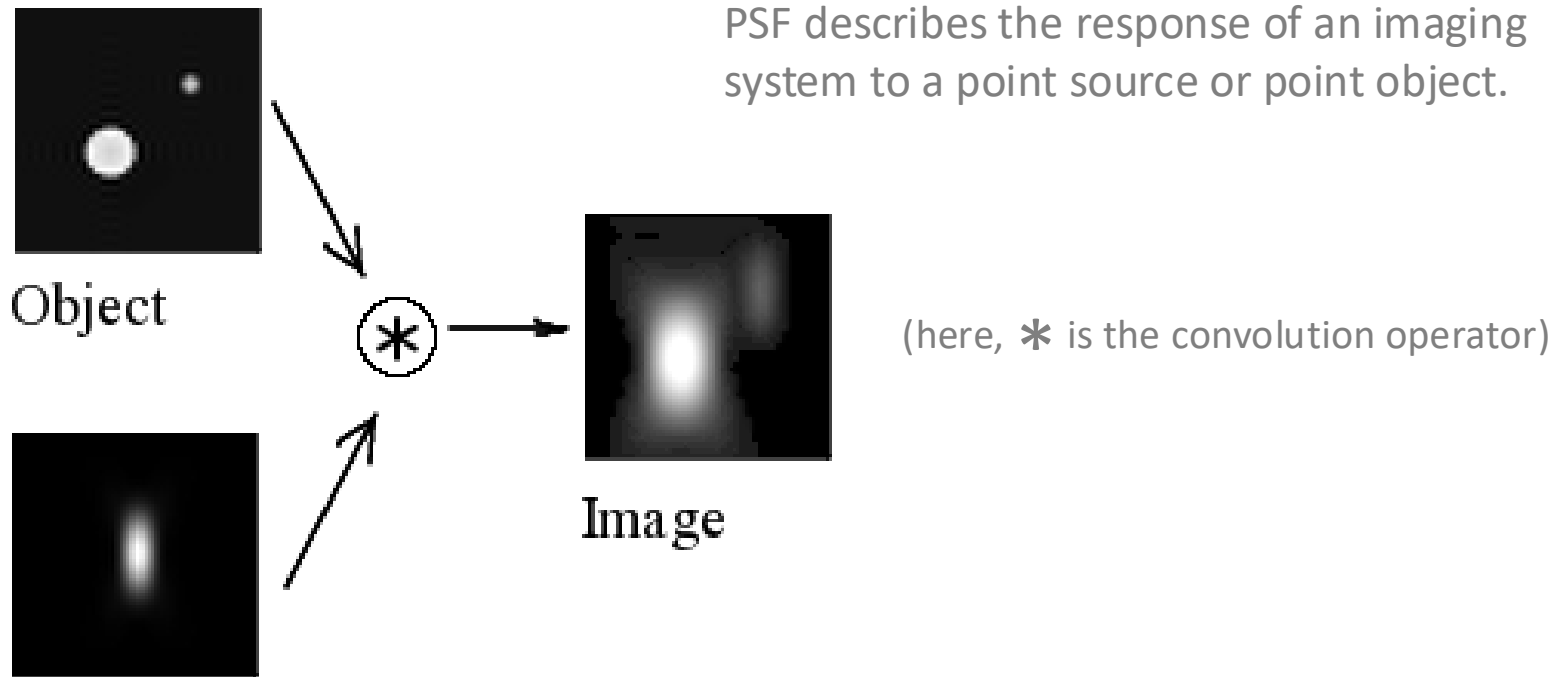
Rayleigh criterion



A generally accepted definition of the resolution of a diffraction-limited image:

The first diffraction minimum of the image of one source point coincides with the central maximum of another.

Point spread function (PSF)

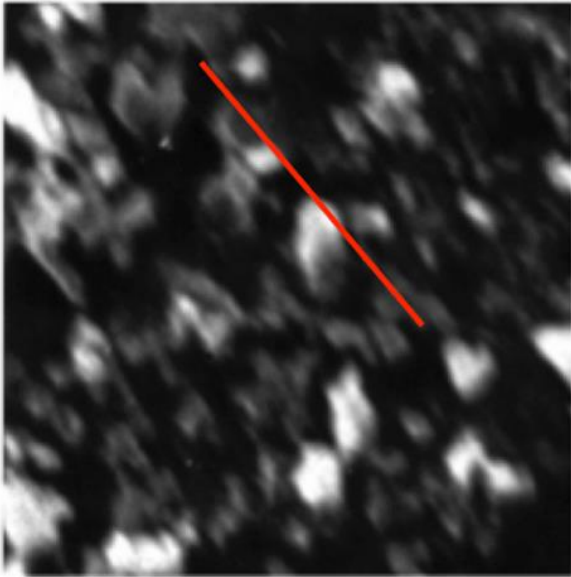


PSF

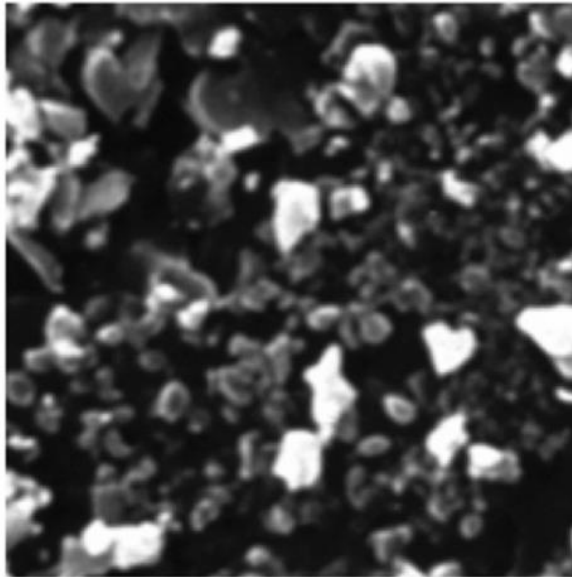
e.g., cross-sectional “shape” of the electron beam used in a scanning electron microscope

Useful concept in 2D (and 3D) microscopy as it describes all aspects of resolution in 2D (or 3D), as shown above by a 2D *astigmatic* PSF. Astigmatism is a type of aberration that makes the resolution of an imaging system anisotropic. It therefore makes the 2D PSF to be non-circular, and gives rise to geometric distortions in images.

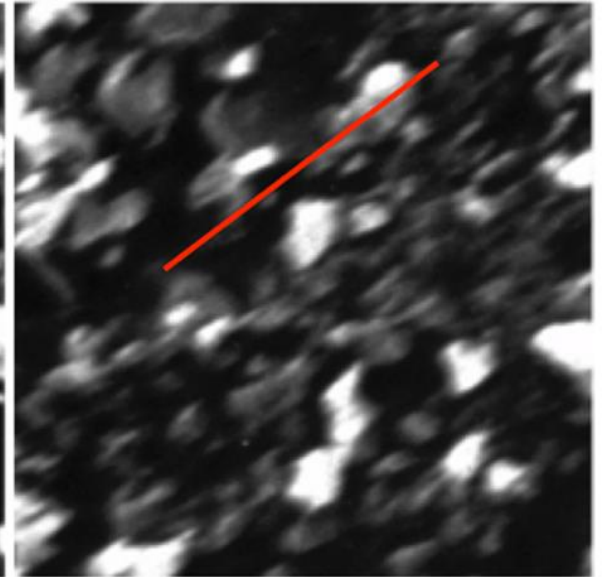
Astigmatism aberration



resolution is different in different directions



resolution limit seen as blurring of the image



resolution is different in different directions

What's next?

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Thurs 31 July 1 PM: Prac 0: introduction to electron & ion beam microscopy
 Friday 1 Aug 3 PM: Tute: ImageJ image processing